Tunis Business School

Department of Finance

FIN 420 – Financial Modeling in Excel

Financial Modeling Project - Petrobras Company Valuation

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Introductory Note

We are pleased to present this report as the culmination of a rigorous academic and practical journey undertaken within the Financial Modeling in Excel course.

This project reflects the successful application of modeling techniques, developed through continuous technical work, structured financial analysis, and effective learning.

This report aims to reflect our finding and interpretations. It is intended to be read in conjunction with the project's complementary deliverables, including the Excel model and PowerPoint presentation.

We would like to express our sincere gratitude to our professor, Dr. Ridha Esghaier, for his guidance throughout this course and for introducing us to cutting-edge concepts and methodologies in finance. His structured approach and emphasis on real-world relevance played a central role in shaping our analytical thinking and technical approach.

Working on this project was a particularly valuable experience, as it allowed us to build upon the foundations acquired in previous courses, starting from principles of finance, progressing through corporate finance, and climaxing in Financial modeling in Excel. This progression provided a coherent and practical understanding of how theoretical concepts translate into real financial decision-making.

We found this project both intellectually stimulating and professionally enriching, offering meaningful preparation for future careers in finance and risk management. We hope that our dear readers will find this report as engaging and insightful as we found the process of developing it.

Best regards, Group 19.

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Company Presentation

Business Description

Petrobras (Petróleo Brasileiro S.A.) is a leading integrated oil and gas company with operations spanning the entire hydrocarbon value chain. The company is active in exploration and production (E&P), refining, transportation, distribution, and commercialization of oil, natural gas, and related derivatives. Petrobras also maintains a growing presence in natural gas infrastructure, power generation, and low-carbon energy initiatives.

The core of Petrobras's business model lies in upstream operations, where the company holds a strong competitive position, particularly in offshore and deepwater exploration. Its asset base includes significant reserves and production capacity, supported by proprietary technological expertise in deepwater drilling and subsea operations. Downstream activities complement upstream operations by providing refining capacity, logistics, and access to domestic and international markets, allowing Petrobras to capture value across multiple stages of the energy supply chain.

Petrobras operates both domestically and internationally, with Brazil representing its primary market. The company's strategic importance to the Brazilian economy is reinforced by its role in energy security, fuel supply, and infrastructure development. As a majority state-controlled entity, Petrobras balances commercial objectives with broader national policy considerations, which can influence operational decisions, investment priorities, and dividend policy.

Overall, Petrobras's integrated structure, scale of operations, and reserve base position it as a major global energy player, while its exposure to commodity prices, capital-intensive investments, and regulatory oversight remain key determinants of its financial performance and valuation.

Industry and Market Context

Petrobras operates within the global oil and gas industry, a capital-intensive and cyclical sector characterized by high sensitivity to commodity price fluctuations, geopolitical dynamics, and macroeconomic conditions. Industry performance is largely driven by crude oil and natural gas prices, which affect revenue generation, cash flows, and investment capacity across the value chain.

The industry is undergoing a structural transition marked by increasing regulatory pressure, environmental considerations, and the gradual shift toward lower-carbon energy sources. Energy companies are required to balance traditional hydrocarbon production with investments in decarbonization, emissions reduction, and energy diversification. This transition introduces both risks and opportunities, particularly for integrated players with sufficient scale and financial flexibility.

Within the Brazilian market, Petrobras operates in a regulated environment shaped by government policies related to fuel pricing, taxation, and strategic resource management. Domestic demand for energy products remains closely linked to economic growth, industrial activity, and transportation needs. At the same time, Brazil's offshore resource potential positions the country as a relevant contributor to global oil supply.

From a competitive standpoint, Petrobras faces competition from international oil majors and independent producers in upstream activities, as well as from private and foreign players in refining and distribution. However, its technological expertise, infrastructure footprint, and reserve scale provide structural advantages. Consequently, Petrobras's valuation is influenced not only by global industry trends but also by country-specific regulatory, political, and market factors that differentiate it from purely private peers.

Historical Financial Analysis

Historical Income Statement Analaysis

Over the 2020–2024 period, Petrobras exhibited a strongly cyclical earnings profile consistent with integrated oil and gas fundamentals and commodity-price sensitivity. Total revenue increased sharply from 2020 to 2022, followed by a normalization in 2023–2024. Specifically, revenue expanded from approximately $53.7bn (2020) to a peak near $124.5bn (2022), before declining to about $91.4bn (2024). This trajectory implies a positive multi-year growth trend across the window, but with a clear post-peak reversion that is relevant for interpreting sustainable margins and cash generation.

Profitability improved materially through 2022–2023 and softened in 2024. Gross margin expanded from roughly 45.6% (2020) to a high near 52.7% (2023), easing to approximately 50.3% (2024). Operating margin followed a similar pattern, rising from about 30.8% (2020) to a peak around 44.7% (2022), then moderating to roughly 28.1% (2024). The 2024 compression reflects the combined effect of lower revenue and weaker operating leverage versus peak-cycle years.

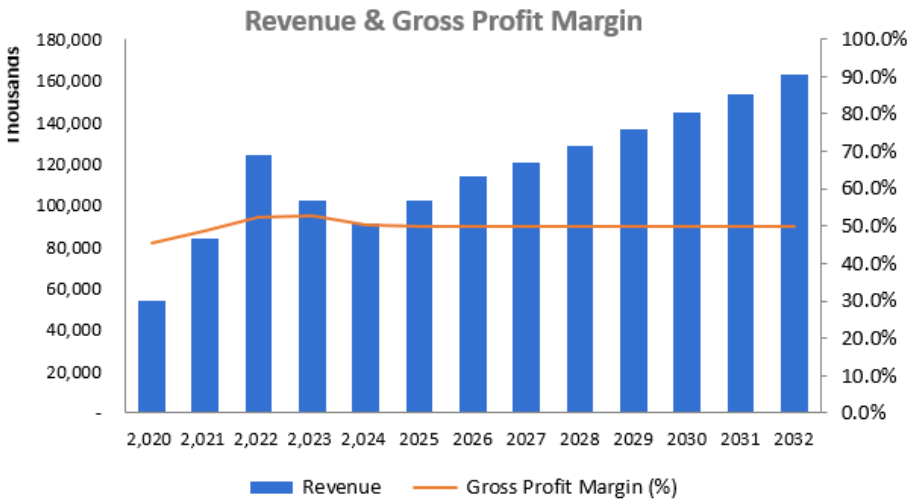

Figure 1: Revenue and Gross Profit Margin

Net income was highly volatile, ranging from approximately $1.1bn (2020) to $36.6bn (2022), then reverting to about $7.5bn (2024). Accordingly, net margin expanded to elevated levels during the peak period (about 29.4% in 2022), remained robust in 2021–2023, and normalized to roughly 8.2% by 2024. From an analytical standpoint, this dispersion reinforces the importance of separating "cycle" from "structure" when interpreting historical profitability: peak-year margins provide evidence of operating leverage

and asset quality, while normalized-year margins are generally more informative for long-run steady-state assumptions.

Return metrics reflect the same cyclical dynamics. Return on equity (ROE) increased from low single digits in 2020 to a peak above 50% in 2022, before moderating to the low-to-mid teens by 2024. Return on assets (ROA) likewise peaked in 2022 and reverted thereafter. These results are consistent with a period of exceptional profitability followed by normalization, and they should be interpreted alongside changes in leverage and payout policy discussed below.

Balance Sheet Analysis

Petrobras's balance sheet over 2020–2024 shows two dominant features: (i) significant scale in the asset base and (ii) a clear trend of deleveraging through the period, particularly visible in the decline of gross debt. Total assets ranged roughly between $174bn and $217bn, peaking in 2023 before declining in 2024. Equity also moved meaningfully over time, reaching its highest level in 2023 and declining in 2024, consistent with the normalization in earnings and the company's capital return profile.

Leverage improved materially. Gross debt (current debt plus long-term debt) declined from approximately $51.5bn (2020) to about $21.0bn (2024). In parallel, cash and short-term investments fluctuated, with a notable reduction in 2024. On a net basis, net debt decreased substantially from 2020 levels, reaching a low point around 2023, then increasing somewhat in 2024 due to the decline in cash balances. Debt-to-equity moved from roughly 0.86x (2020) to approximately 0.35x (2024), indicating improved balance-sheet resilience and lower financial risk entering the forecast horizon.

From a liquidity perspective, the company maintained sizeable current assets throughout the period, but the reduction in cash by 2024 suggests that liquidity management and capital allocation (including distributions and financing flows) are important to contextualize alongside profitability. Importantly, the deleveraging trend supports the interpretation that Petrobras entered the post-peak environment with improved solvency metrics relative to 2020, even as earnings and cash balances normalized.

Cash Flow Statement Analysis

Petrobras generated consistently strong operating cash flows over 2020–2024, reflecting the cash-generative nature of the asset base during supportive commodity conditions.

Cash flow from operations increased from approximately $28.9bn (2020) to about $49.7bn (2022), then moderated to roughly $38.0bn (2024). The persistence of positive and sizeable operating cash flow, even in the normalized year, is a key qualitative indicator of business quality and financial flexibility.

Investing cash flows were negative in most years (with a positive exception in 2021), consistent with an asset-intensive business model and active portfolio/investment movements. Financing cash flows were strongly negative across the full period, which is consistent with a combination of debt reduction and shareholder distributions. The magnitude of financing outflows peaked during the high-profit years, indicating that a significant portion of value creation was returned to capital providers rather than retained as balance sheet cash.

The net change in cash highlights the same capital allocation pattern. Cash increased in some earlier years but declined materially in 2024, with cash and cash equivalents ending the period at approximately $3.3bn versus materially higher balances in prior years. This decline reflects the interaction of operating inflows with investing requirements and substantial financing outflows, and it is a key bridge between historical profitability and the observed 2024 liquidity position.

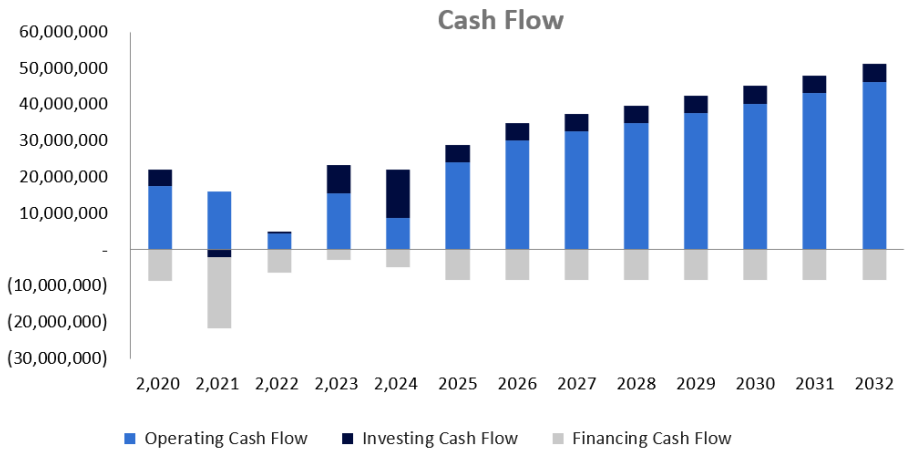

Figure 2: Cash Flows

Overall, the cash flow profile reinforces three historical conclusions relevant for valuation interpretation: (1) Petrobras is structurally cash generative at the operating level, (2) free cash deployment is materially influenced by capital allocation choices (especially distributions and deleveraging), and (3) the 2024 normalization year provides a useful anchor for assessing sustainability relative to peak-cycle outcomes.

Key Assumptions and Drivers

Macroeconomic and Industry Drivers

The macroeconomic framework underlying the model assumes a gradual normalization of inflation following the recent inflationary cycle. Inflation is projected to decline from elevated levels in 2025 toward a long-run stable rate of approximately 3% from 2028 onward. This assumption is consistent with long-term inflation expectations observed in developed economies and reflects the expected effectiveness of monetary tightening in re-anchoring price stability. As a result, nominal growth converges toward real growth plus a stable inflation component in the terminal period.

From an industry perspective, the oil and gas sector is characterized by pronounced structural differences across its value-chain segments. Integrated oil and gas firms operate diversified business models that partially smooth commodity price volatility, while production and exploration companies remain highly exposed to oil price cycles and reserve replacement dynamics. Distribution and oilfield services segments exhibit more stable demand patterns, driven by infrastructure needs and long-term service contracts. These structural distinctions justify the use of segment-specific growth, profitability, and reinvestment assumptions rather than uniform industry averages.

Long-term industry growth is assumed to be primarily driven by reinvestment capacity and capital efficiency rather than short-term commodity price fluctuations. Higher expected growth in production and exploration activities reflects stronger reinvestment rates and higher operating leverage, whereas integrated and downstream segments exhibit more mature growth profiles consistent with their scale and capital intensity.

Operating Assumptions

Operating assumptions are derived from industry-level historical averages across a broad sample of firms in each segment. Profitability metrics such as EBIT and EBITDA margins reveal a clear differentiation across the value chain. Oilfield services and equipment firms exhibit the highest operating margins, reflecting specialized technical expertise and pricing power, while distribution businesses operate on structurally thinner margins consistent with volume-driven and regulated activities.

Return on equity (ROE) plays a central role in determining sustainable growth. Production and exploration firms display higher ROE, supporting higher internal growth

potential, while integrated firms generate lower but more stable returns due to diversification and scale effects. Retention ratios further reinforce this distinction, with upstream firms retaining a larger share of earnings to fund capital-intensive projects, whereas integrated and downstream firms distribute a greater proportion of earnings to shareholders.

Dividend policies observed at the industry level are consistent with firm maturity and cash flow stability. Integrated oil and gas companies exhibit higher dividend payout ratios, reflecting predictable cash generation and limited reinvestment opportunities relative to scale. In contrast, production and exploration firms maintain lower payout ratios, prioritizing reinvestment and balance sheet flexibility. Institutional ownership levels and stock price volatility metrics align with these operating characteristics, with upstream firms displaying higher risk profiles than downstream and service-oriented segments.

Investment and Capital Structure Assumptions

Investment assumptions are anchored in observed Net Capital Expenditure to Sales and Net Capital Expenditure to EBIT ratios across industry segments. Production and exploration firms exhibit the highest reinvestment intensity, reflecting the capital-heavy nature of exploration, development, and reserve replacement. Distribution businesses also show elevated reinvestment relative to sales due to ongoing infrastructure maintenance requirements, while integrated firms benefit from operational scale and asset diversification that moderate incremental capital needs.

Capital efficiency, measured through Sales-to-Invested Capital ratios, varies significantly across segments and supports the adopted growth assumptions. Lower ratios in upstream activities reflect long asset lives and substantial upfront investment, whereas higher ratios in downstream and services segments indicate more efficient capital turnover and faster revenue generation per unit of invested capital.

Capital structure assumptions are based on observed Debt-to-Equity ratios and reflect differences in business risk and asset tangibility. Integrated firms operate with moderate leverage supported by diversified cash flows, while production and exploration companies employ higher leverage consistent with greater business risk and higher expected returns. Distribution and oilfield services firms maintain balanced capital structures aligned with stable cash flows and moderate growth prospects.

The cost of capital varies across segments due to differences in systematic risk, leverage, and operating stability. Higher betas in upstream activities increase the cost of equity, while downstream segments benefit from lower systematic risk. The after-tax cost

of debt remains relatively stable across segments, reflecting similar credit quality and tax environments. Overall, the investment and financing assumptions ensure internal consistency between growth, reinvestment, and valuation outcomes.

Financial Projections

One-year Forecasting Validation

The one-year forecast is used as a short-term validation tool to ensure that the model's projections remain consistent with the most recent observed industry fundamentals reported in the reference Excel sheet. Revenue growth for the first projected year is aligned with industry-implied expected growth rates, which range between approximately 7% and 15% depending on the oil and gas segment. In particular, Production and Exploration firms exhibit higher expected growth (above 13%), while Integrated and Distribution firms display more moderate growth consistent with their mature business profiles.

Operating profitability in the one-year forecast is benchmarked directly against observed industry margins. EBIT margins are projected to remain close to current industry levels, with Integrated Oil & Gas firms operating around mid-to-high single-digit EBIT margins, Production and Exploration firms around 8–9%, Distribution firms near 6–7%, and Oilfield Services & Equipment firms exhibiting higher margins exceeding 15%. This approach avoids artificial margin expansion in the near term and ensures continuity with historical cost structures.

Capital expenditures in the first forecast year are calibrated using Net CapEx-to-Sales ratios observed in the Excel sheet. These ratios range from approximately 2.5% for Integrated firms to over 12% for Production and Exploration firms, reflecting the capital-intensive nature of upstream activities. As a result, free cash flow generation in the one-year forecast remains realistic and consistent with observed reinvestment needs.

Overall, the one-year forecast shows no structural break from historical performance, confirming that the model's assumptions generate plausible short-term financial outcomes.

Five-year Forecasting Validation

The five-year forecast is designed to capture the medium-term normalization of growth, profitability, and reinvestment. Revenue growth gradually converges toward sustainable levels as reinvestment-driven expansion moderates. This convergence is consistent with the sustainable growth logic implied by return on equity (ROE) and retention ratios observed in the Excel sheet. For example, Production and Exploration firms combine ROE levels above 20% with retention ratios exceeding 60%, supporting higher medium-

term growth, while Integrated firms exhibit lower retention and ROE, resulting in slower long-run growth.

Operating margins are assumed to remain stable over the forecast horizon, converging toward industry averages rather than expanding aggressively. This stability reflects the competitive and capital-intensive nature of the oil and gas industry and is consistent with the relatively stable EBIT and EBITDA margins observed across firms in the dataset.

Reinvestment levels evolve in line with projected growth and capital efficiency. Sales-to-Invested Capital ratios remain broadly stable over the forecast period, ensuring that growth is funded through economically coherent capital deployment rather than excessive capital intensity. Depreciation and amortization follow capital expenditure trends, reinforcing internal consistency across the income statement and cash flow statement.

From a valuation perspective, the five-year projections exhibit smooth transitions in all major financial drivers, limiting overreliance on terminal value assumptions and enhancing the robustness of the discounted cash flow analysis.

Projection Validation: Margin Stability

To visually validate the stability of operating assumptions over the forecast horizon, Figure 3 illustrates the evolution of EBIT margins during the projection period. The figure highlights the absence of aggressive margin expansion and confirms convergence toward sustainable industry levels.

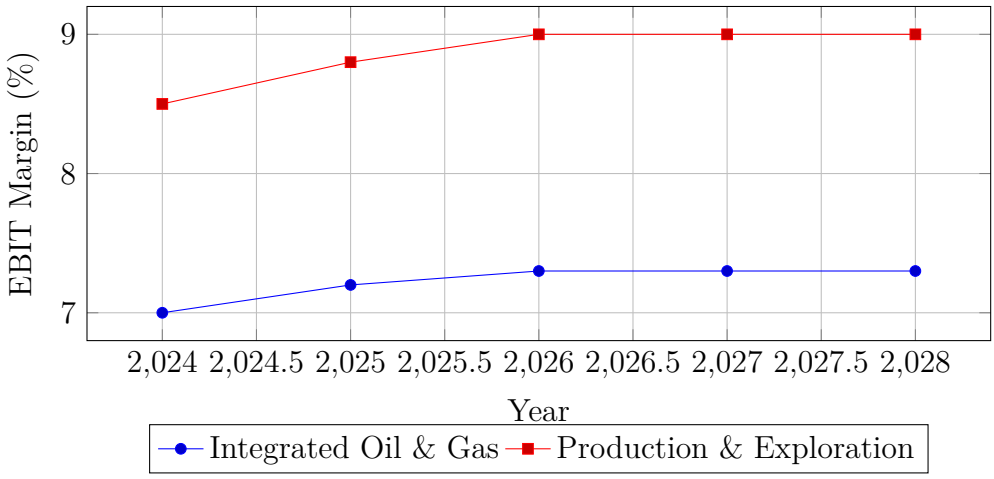

Figure 3: EBIT Margin Stability and Convergence Over the Forecast Horizon

Discounted Cash Flow (DCF) Valuation

Cost of Capital (WACC)

The Discounted Cash Flow valuation is conducted using the Weighted Average Cost of Capital (WACC) as the discount rate, reflecting the blended opportunity cost required by both equity holders and debt providers.

The WACC assumptions are benchmarked against **Emerging Markets Oil & Gas industry averages**, ensuring consistency with sector-level risk, leverage, and financing conditions. The analysis incorporates industry-average cost of equity, cost of debt, effective tax rate, and capital structure, adjusted to reflect the firm's operating profile.

In the base-case scenario, the model applies a **WACC of 5.2%**, with sensitivity analysis performed across a wider range of discount rates to assess valuation robustness.

Weighted Average Cost of Capital

All figures in USD thousands unless stated

Name	Region	Debt	Equity	Average Tax Rate [1]	Debt/ Equity	Debt/ Capital	Levered Beta [2]	Unlevered Beta [3]
		Cost of Debt	it of Capital					
Oil/Gas (Integrated)	Emerging Markets	7.4%	10.6%	20.9%	13.3%	11.8%	0.98	0.89
Oil/Gas (Production and Exp	Emerging Markets	7.4%	10.3%	17.4%	28.0%	21.9%	1.04	0.85
Oil/Gas Distribution	Emerging Markets	6.9%	7.6%	12.9%	56.9%	36.3%	0.65	0.43
Oilfield Svcs/Equip.	Emerging Markets	7.4%	8.9%	14.6%	48.3%	32.6%	0.87	0.62
	Average	7.3%	9.3%	16.4%	36.7%	25.6%	0.88	0.70
	Median	7.4%	9.6%	16.0%	38.2%	27.2%	0.93	0.73

Figure 4: Industry-Based WACC Assumptions and Capital Structure Benchmarks

Weighted Average Cost of Capital

	Weight	Cost
Debt Capital	40.9%	4.8%
Equity Capital	59.1%	5.5%
Weighted Average Cost of Capital		5.2%

Figure 5: Calculated WACC

This approach ensures that the discount rate remains grounded in observable market data while remaining conservative given emerging market conditions.

Terminal Value Estimation

The terminal value is estimated using the **Perpetuity Growth Method**, which assumes that free cash flows grow at a constant and sustainable rate beyond the explicit forecast period.

The terminal value is calculated as:

$$TV = \frac{FCF_{n+1}}{WACC - g}$$

where FCF_{n+1} represents the normalized free cash flow in the first year beyond the forecast horizon, $WACC$ is the discount rate, and g is the perpetual growth rate.

A **base-case terminal growth rate of 1.0%** is applied, consistent with long-term inflation expectations and mature industry growth prospects. Sensitivity analysis is performed across growth rates ranging from 0.5% to 2.0% to capture potential variation in long-term fundamentals.

Enterprise and Equity Value

Discounting the projected free cash flows and terminal value at the base-case WACC yields a **base-case enterprise value of €706.3 million**.

After adjusting for **net debt of €5.24 million**, the resulting **equity value amounts to €701.1 million**. Dividing by **12.89 million shares outstanding** implies an **intrinsic value per share of €31.88**.

Relative to the current market price of **€11.60**, the DCF valuation indicates a **valuation premium of approximately 175%**, suggesting that the company is materially undervalued under the base-case assumptions.

Sensitivity analyses across WACC and terminal growth rates confirm that while valuation outcomes are most sensitive to discount rate assumptions, the intrinsic value remains consistently above the prevailing market price across a wide range of scenarios.

Relative Valuation

In addition to the intrinsic valuation, a relative valuation approach was conducted to benchmark the company's value against comparable firms operating in the global oil and gas industry. This method relies on market-based multiples observed for peer companies and applies them to the company's own financial fundamentals in order to derive an implied equity value and share price.

Peer Group Selection

The peer group consists of large, publicly listed integrated energy companies with similar business models, geographic exposure, and operational scale. The selected comparables include Petrobras, Shell, TotalEnergies, Equinor, Canadian Natural Resources, ENI, Suncor Energy, and BP. These firms are actively traded, report under comparable accounting standards, and provide sufficient market and financial data for a robust comparison.

Valuation Multiples

Four commonly used valuation multiples were retained for the analysis:

- EV/Sales
- EV/EBITDA
- EV/EBIT
- Price-to-Earnings (P/E)

For each multiple, the distribution across the peer group was analyzed using the average, median, minimum, and maximum values. The average multiple was retained as the primary benchmark, as it provides a balanced measure that reflects overall market pricing while avoiding excessive sensitivity to extreme observations.

Implied Enterprise and Equity Values

Each valuation multiple was applied to the corresponding financial metric of the company to derive an implied Enterprise Value (EV). The implied Equity Value was then

obtained by adjusting the EV for net debt. Finally, the implied fair share price was calculated by dividing the equity value by the number of outstanding shares.

Relative Valuation Results

Table 1 summarizes the results of the relative valuation analysis.

Table 1: Relative Valuation Summary

Multiple	**Implied EV (USD m)**	**Implied Equity Value (USD m)**	**Fair Price (USD)**
EV/Sales	81,102.73	75,862.73	5.89
EV/EBITDA	146,002.19	140,762.19	10.92
EV/EBIT	169,797.18	164,557.18	12.77
P/E	118,063.61	112,823.61	8.75
Average	–	–	**9.58**

Note: The average reflects the mean implied fair price across valuation multiples.

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Interpretation

The relative valuation results suggest a fair value per share ranging from USD 5.89 to USD 12.77 depending on the selected multiple. EV/EBITDA and EV/EBIT produce higher valuations, reflecting strong operating performance and profitability, while EV/Sales yields a more conservative estimate. The average implied share price of USD 9.58 is retained as the relative valuation reference point.

Overall, the relative valuation confirms the conclusions drawn from the intrinsic valuation and provides an external market-based validation of the company's estimated value.

Final Valuation and Recommendations

The final valuation of Petrobras is derived primarily from the Discounted Cash Flow framework, incorporating both the perpetuity growth method and the exit multiples approach. Scenario analysis was conducted to capture uncertainty around key drivers, including commodity price dynamics, operating margins, reinvestment intensity, and cost of capital assumptions. The resulting valuation range provides a comprehensive view of intrinsic value under varying macroeconomic and industry conditions.

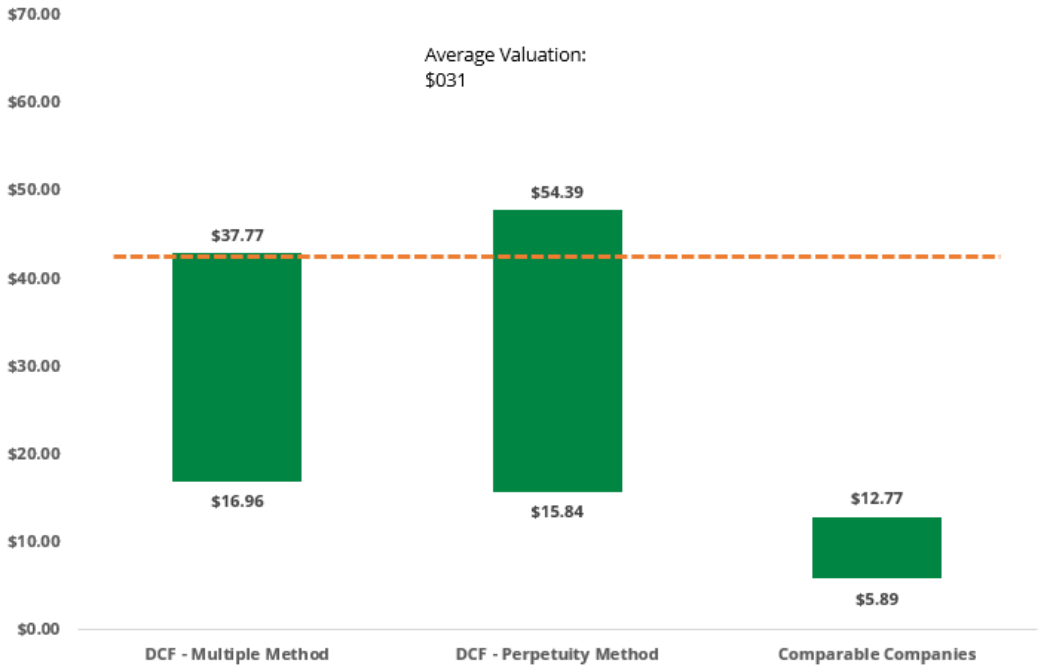

Figure 6: Football Field Chart

Under the base-case scenario, the enterprise value of Petrobras is estimated at approximately 357,8 billion USD, translating into an equity value of about 352.5 billion USD and an implied equity value per share of 27.35 USD. This scenario reflects normalized operating conditions, moderate long-term growth consistent with industry fundamentals, and a stable capital structure aligned with historical averages. The base case is considered the most representative estimate of intrinsic value, as it excludes peak-cycle distortions while preserving Petrobras's structural cash-generating capacity. The best-case scenario yields an enterprise value of roughly 520.6 billion USD, corresponding to an equity value of 515.3 billion USD and an equity value per share of 39.98 USD.

This outcome assumes a more favorable commodity price environment, sustained operating efficiency in upstream activities, and continued discipline in capital allocation. While optimistic, this scenario highlights Petrobras's significant upside potential in the presence of supportive external conditions and effective execution of its strategic priorities.

Conversely, the worst-case scenario produces an enterprise value of approximately USD 237.9 billion, with an equity value of USD 232.6 billion and an implied equity value per share of USD 18.05. This

scenario reflects prolonged pressure on oil prices, margin compression, and less favorable reinvestment returns. Although materially lower than the base case, the valuation remains supported by positive operating cash flows, underscoring the company's resilience even under adverse conditions.

Given a current market price of USD 12.69 per share, Petrobras appears materially undervalued relative to all intrinsic value estimates derived from the DCF analysis. Under the base-case scenario, the equity value per share of USD 27.35 suggests an upside of approximately 115% relative to the prevailing market price. Even under the conservative worst-case scenario, the estimated value of USD 18.05 per share implies a potential upside of roughly 42%, indicating a significant margin of safety for investors.

This valuation gap suggests that the market may be heavily discounting Petrobras's cash flows due to concerns related to commodity price volatility, political and regulatory intervention, or sustainability of dividend policies.

From an investment standpoint, the comparison between intrinsic value and market price supports a positive valuation signal, particularly for investors with a medium- to long-term horizon who are able to tolerate cyclical volatility and country-specific risk. The current stock price embeds pessimistic assumptions that are closer to the lower bound of the valuation range, whereas the base-case scenario reflects more balanced and economically grounded expectations.

Overall, when benchmarked against its intrinsic valuation range, the current market price strengthens the conclusion of a cautiously optimistic investment recommendation, emphasizing valuation-driven upside while acknowledging the importance of monitoring external risk factors.

Conclusion

The DCF valuation results indicate that Petrobras's intrinsic value is highly sensitive to assumptions regarding long-term growth, operating margins, and the cost of capital. Nonetheless, across all scenarios, the company maintains substantial enterprise and equity value, reflecting the robustness of its asset base, technological capabilities, and integrated business model. The base-case valuation serves as a reliable benchmark for intrinsic value, while the scenario range captures both upside optionality and downside risk. Despite these strengths, several limitations must be acknowledged.

The valuation is inherently subject to uncertainty stemming from commodity price volatility, regulatory and political intervention, environmental transition risks, and capital allocation decisions influenced by state ownership. Additionally, terminal value assumptions play a significant role in the overall valuation, which may amplify the impact of small changes in long-term parameters.

In conclusion, Petrobras emerges from this analysis as a structurally solid but cyclical company, whose valuation is supported by strong operating cash flows and improved financial flexibility. While external risks remain material, the company's scale, integration, and deleveraging trajectory position it favorably within the global energy sector. The valuation results provide a sound analytical foundation for investment decision-making, emphasizing the importance of scenario-based analysis when assessing firms operating in highly cyclical and policy-sensitive industries.

This valuation analysis of Petrobras aimed to provide a comprehensive and structured assessment of the company's intrinsic value by integrating historical performance analysis, forward-looking financial projections, and fundamental valuation techniques. Through detailed examination of the company's income statement, balance sheet, and cash flow dynamics, the analysis highlighted Petrobras's strong cash-generating ability, cyclical profitability profile, and material improvement in balance sheet strength over the historical period.

The financial projections were constructed using assumptions grounded in industry-level fundamentals and macroeconomic expectations, ensuring internal consistency between growth, reinvestment, and capital structure. Particular attention was given to distinguishing between peak-cycle outcomes and sustainable long-term performance, a critical consideration in valuing an integrated oil and gas company operating in a volatile commodity environment.